

Task: Expose the minikube cluster dashboard to be accessed from a browser outside the cluster.

Steps:

1- create a **NodePort service** with a specific port to access the cluster using this port
The yaml file for the NodePort service:

```
apiVersion: v1
kind: Service
metadata:
  name: kubernetes-dashboard-nodeport
  namespace: kubernetes-dashboard # the namespace of the dashboard
spec:
  type: NodePort
  selector:
    k8s-app: kubernetes-dashboard # the label of the dashboard deploy
    or app
  ports:
    - port: 80
      targetPort: 9090 # the port used by the minikube dashboard inside the cluster
      nodePort: 30050 # specify any available port between 30000 and 32767
```

2- **kubectl apply -f dashboard-svc.yml** -----> to apply the file for creating the service

3- **kubectl get svc -n kubernetes-dashboard** -----> to make sure that the service is created successfully

3- **minikube ip** -----> to show the minikube ip

4- open the nginx configuration file:

vim /etc/nginx/nginx.conf

then write then add the following block:

```
server {
    listen 8150 ;
    listen [::]:8150 ;
    server_name dashboard-service;

    include /etc/nginx/default.d/*.conf;
```

```

    error_page 404 /404.html;
    location / {
        proxy_pass http://192.168.49.2:30050;
# this is the ip of the minikube cluster and the port of the dashboard nodeport service
    }

    error_page 500 502 503 504 /50x.html;
    location = /50x.html {
    }
}

```

5- `sudo nginx -t` -----> to ensure that the nginx conf file syntax is valid

6- `sudo systemctl reload nginx.service` -----> to make the system see the new configuration edits of the nginx service

7- go to the browser and write the ip of the server or machine and the port of the dashboard service like this:
10.100.15.25:30050